Concept Guide: Smart Content Summarizer

1. Project Overview

Smart Content Summarizer (SummaSmart) is a workflow-based application built using Google Opal. It accepts a user's summarization instructions and an article URL, retrieves webpage content, generates a summary, and presents the result in a readable Markdown webpage.

2. Problem Statement

Online articles can be lengthy and time-consuming to read. Users may need a shorter version that highlights the main ideas while following specific requirements for length, format, language, and tone.

3. Project Objective

The objective is to demonstrate how Google Opal can orchestrate user inputs, AI reasoning, a webpage retrieval tool, and a final output step in a single workflow.

4. Target Users

- Students reviewing online learning materials.
- Professionals reviewing articles and reports.
- Researchers exploring publicly accessible web content.
- General users who need concise article summaries.

5. Main Features

1. Collects custom summarization instructions.
2. Collects an article URL.
3. Retrieves webpage content using the connected Get Webpage tool.
4. Summarizes retrieved content according to the user's requirements.
5. Displays the instructions, source URL, summary, and original content.
6. Uses Markdown headings, lists, and clickable links for readability.

6. Workflow Architecture

The workflow contains five main steps:

1. **User Instructions:** Collects the user's preferred summary requirements.
2. **Get Content Source:** Collects the URL of the target article.
3. **Fetch content:** Calls the connected Get Webpage tool to retrieve webpage text.
4. **Generate Content Summary:** Uses the retrieved content and user instructions to generate the summary.
5. **Render Summary Webpage:** Displays the source, instructions, summary, and original content.

7. Input and Output

Inputs

- User summarization instructions.
- Article URL.

Outputs

- Original user instructions.
- Clickable source URL.
- Generated summary.
- Retrieved article content.

8. Accuracy and Responsible Use

- Use retrieved article content as the factual source.
- Do not invent facts when retrieval fails.
- Respect the user's requested format and length where feasible.
- Clearly indicate when webpage content cannot be retrieved.
- Do not assume that every webpage is accessible to the retrieval tool.
- Users should verify important information against the original source.

9. Scope and Limitations

The application focuses on retrieving and summarizing accessible webpage text. It does not guarantee access to login-protected, blocked, dynamically rendered, or otherwise inaccessible pages. It does not guarantee perfect factual accuracy or provide a substitute for reviewing the original article.

The interface uses Markdown. Advanced features such as interactive drawers, animated transitions, and floating buttons depend on the capabilities of the platform.

10. Expected Outcome

A working Google Opal application that demonstrates a multi-step workflow, webpage retrieval, instruction-based summarization, and a formatted final report.